

FedShield-ID

Privacy-First Identity Trust & Risk-Based Adaptive Authentication Platform for Banking Networks

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Target Audience	Banking SOC Teams, Security Officers, Compliance Auditors
Core Concepts	Post-Quantum Cryptography, Differential Privacy, Federated Learning, Behavioral Biometrics, Shapley Attribution (XAI)

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1. Executive Summary

Modern banking security is shifting from basic transaction-level fraud checks to **continuous identity trust validation**. Legacy architectures fail to detect sophisticated synthetic identities, hijacked sessions, and insider threat data exports. Furthermore, modern centralized machine learning systems create immense privacy liabilities, pooling raw financial data and violating strict regulatory mandates.

FedShield-ID solves these challenges through a decentralized, privacy-preserving identity trust framework. By integrating behavioral analytics, device reputation metrics, and post-quantum security, the platform continually computes an active **Identity Trust Score (0-100)** for every session. Instead of creating massive data warehouses, FedShield-ID trains machine learning models locally at distinct banking nodes and aggregates them securely via Privacy-Preserving Federated Learning and Differential Privacy.

2. System Architecture

The FedShield-ID system operates as a distributed multi-layered pipeline across three independent simulated bank nodes (Bank A: Retail, Bank B: Premium Cards, and Bank C: Savings). The trust verification workflow follows a strict path from onboarding checks to decentralized compliance ledger indexing:

- **Onboarding Guardrails:** Format checking, disposable domain filters, and synthetic identity score calculation.
- **Continuous Telemetry Ingestion:** Tracking keyboard/mouse dynamics alongside device identifiers and network nodes.
- **Composite Risk Computation:** Trust Scoring Engine evaluating 10 distinct scoring dimensions in real-time.
- **Adaptive Response (Auth Decisions):** Grading friction levels (Frictionless, OTP, Step-up, Biometric Face, or Hard Block).
- **Decentralized Aggregation:** Banks update the global model collaboratively via Federated Averaging (FedAvg).
- **Privacy & Encryption:** Laplacian noise addition (Differential Privacy) and simulated CRYSTALS-Kyber-768 key encapsulation.
- **Explainable Audit Trail:** Shapley values mapping exact risk attributions with GenAI natural language briefings.

3. Detailed Feature Walkthrough

3.1 Onboarding Verification & Synthetic ID Prevention

First-line checks prevent synthetic accounts at creation. The system validates standard Permanent Account Number (PAN) formats, scans email addresses against known disposable/temporary email domains, and triggers VoIP carrier detection flags. Any format mismatch or suspicious domain patterns directly reduce the initial *Identity Confidence Score*.

3.2 Continuous Identity Trust Scoring Engine

The platform aggregates 10 distinct telemetry parameters to compute a composite Trust Score (0-100) and its counterpart Risk Score (100 - Trust):

Telemetry Category	Weight	Description / Penalties
Device Reputation	10%	Audits rooted status, emulators, and unrecognized browser signatures.
Login Consistency	10%	Tracks failed logins; deducts 20% score per failed login window.
Geolocation Consistency	10%	Logarithmic penalty based on geo-distance drift from primary home location.
Transaction Volume Anomaly	10%	Flags transaction spikes relative to the historical average profile.
Behavioral Biometrics	20%	Evaluates keys/min, mouse jitter, and click latency. Identifies bot speeds.
Identity Verification Score	15%	Onboarding check results (PAN mismatch, disposable email flags).
Account Recovery Status	10%	Penalizes score if recovery is requested immediately after SIM modifications.
Insider Threat Score	5%	Deducts trust if administrative activities trigger safety bounds.
Active Session Risk	5%	Active session duration, IP shifts, and concurrent login signals.
Authentication History	5%	Historical failed verification penalties and block events.

3.3 Behavioral Biometrics & Bot Detection

FedShield-ID analyzes typing dynamics (keys per minute) and mouse movement jitter (standard deviation offsets). Bots and automated script injections move mouse cursors in perfectly straight lines with zero jitter and perform clicks with sub-millisecond intervals. If click speed is under 0.05 seconds or mouse jitter drops below 0.1, the system triggers a **Robotic Input** signature penalty, immediately slashing the trust score below the critical step-up threshold.

3.4 Adaptive Authentication (Risk-Based Auth)

Verdicts are dynamically mapped to trust boundaries. Scores above 90 enable frictionless access. Scores of 70-90 require a standard SMS/Email OTP. Scores of 50-70 escalate to Step-Up authentication. Scores of 20-50 prompt a Live Face Verification check to audit active biometric identity, and scores below 20 block access entirely.

3.5 Privacy-Preserving Federated Learning & Differential Privacy

Bank nodes train local machine learning classifiers (SGD & Random Forest) on separate SQLite databases. A central aggregator triggers federated communication rounds, combining local model weights using the Federated Averaging (FedAvg) algorithm. Before weight submission, banks apply **Differential Privacy (DP)** by injecting Laplacian noise. This noise addition is mathematically controlled by a privacy budget (epsilon, ϵ). This ensures that a compromised aggregator cannot reconstruct private customer data records from the aggregated model parameters.

3.6 Post-Quantum Security (Simulated Crystals-Kyber)

During federated weight exchange rounds, weights are encrypted with symmetric AES-256-GCM. To negotiate these symmetric keys, the platform simulates the post-quantum CRYSTALS-Kyber-768 Key Encapsulation Mechanism (KEM) over polynomial matrix ring equations ($R_q = \mathbb{Z}_q[X]/(X^{256} + 1)$ with $q = 3329$). The system benchmarks Kyber-768 execution speeds and payload overheads (1184-byte public keys, 1088-byte ciphertexts) against classical RSA-3072 and ECDH-P256, demonstrating resistance against Shor's algorithm.

3.7 Explainable AI & GenAI Forensics

Decisions made by black-box machine learning models are explained using Shapley Additive Explanations (SHAP). The system computes linear SHAP attributions representing risk shifts (e.g., location deviation shifts risk +24.5%). These metrics are passed to an automated GenAI incident briefing agent, which compiles readable briefs outlining identity trust status, primary anomalies detected, and step-by-step mitigation playbooks.

3.8 Identity Risk Ledger & SVG Relationship Graph

All clearances stream to a live regional ledger feed. The relationship analyzer links transaction metadata to construct a relationship knowledge graph. If multiple distinct names share a single device ID or IP address, the visual graph highlights the cluster, enabling SOC teams to detect collusive financial rings.

4. Future Roadmap & Development Plans

To take FedShield-ID from a simulated prototype to a production banking framework, the following enhancements are planned for the future phases of development:

- **True Decentralized Node Deployment:** Migrate from a single-machine simulator to independent Docker clusters running across secure remote channels. Integrate actual gRPC communication channels with Secure Multi-Party Computation (SMPC) layers to coordinate federated rounds between bank nodes.
- **Production Quantum-Safe Integrations:** Transition from simulated CRYSTALS-Kyber logic to native post-quantum cryptographic libraries (such as open-quantum-safe's *liboqs* or OpenSSL 3.0's *oqs-provider*). Build cryptographic fail-safes utilizing hybrid PQC-classical tunnels (e.g., Kyber-ECDH).
- **Browser-Level Biometric Telemetry SDK:** Develop a lightweight JavaScript SDK for bank portals to capture mouse velocity vectors, scroll speed acceleration, keyboard flight times, and touch pressure dynamics. Train local deep learning models on these raw data vectors to build unique biometric identity signatures.
- **Autonomous GenAI Compliance Agent:** Connect the incident briefings to actual Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG). This agent will automatically search banking compliance documents (RBI circulars, KYC guidelines, GDPR frameworks) and auto-draft Suspicious Transaction Reports (STRs) for direct regulatory submission.
- **Real-time Graph Database Integration:** Replace the SQLite-based graph schema with a production graph database (like Neo4j or Amazon Neptune). This will support sub-second query speeds for complex relationship lookups and support automated community-detection algorithms to uncover deep-rooted money laundering rings.

5. Technical Verification & Performance Benchmarks

Verification scripts compare the processing latency of post-quantum operations against standard classical handshakes. While RSA keypairs take significant CPU processing cycles (up to 200ms in python benchmarks), simulated CRYSTALS-Kyber KEM handshakes compute in less than a millisecond. This enables banking networks to enforce quantum-safe data encapsulation without introducing latency overheads in real-time transaction processing pipelines.